A-A-2517
April 1, 1987

COMMERCIAL ITEM DESCRIPTION

BRUSHES, WIRE, ROTARY, WHEEL, TWISTED TUFT

The General Services Administration has authorized the use of this commercial item description in preference to Type III, Class 2 of Federal Specification H-B-771.

This commercial item description covers brushes for use with pneumatic or electric tools in the heavy-duty removal of rust, scale, paint, or burrs from all types of metal surfaces.

Salient characteristics:

Design: Wheel-shape design for use with stationary or portable power tools. Designed to be used individually or ganged on a common arbor. Wire securely attached to the wire-retaining components in a manner that permits the driving torque to be transmitted from the arbor to the brush periphery and that resists the functional forces produced by impact or centrifugal force.

Wire material: High carbon tempered steel or corrosion-resisting steel, as specified in the table.

Wire form: Tufts of U-shaped fill wire passed through or around a retainer and twisted tightly with a left hand helix at least one full turn for brushes less than 10 inches in diameter and at least one and a half turns for brushes 10 and 12 inches in diameter, in a manner that minimizes untwisting during use. Tufts uniform in size, weight, and spacing around the circumference.

Marking: Marked with the Maximum Safe Free Speed (or optionally, MSFS or Maximum RPM), the country of origin, the words "Wear Eye Protection", and the name of the manufacturer or a trademark of such known character that the source of manufacture may be readily determined. Marked speed may be higher than speed listed in table.

Workmanship: The brushes shall be well-constructed with a balanced distribution of wire fill. There shall be no evidence of rust, loose wire, or other defect which may impair serviceability, durability, or appearance.

Standards compliance: The brushes shall comply with American National Standards Institute (ANSI) Standard B165.1, Safety Requirements for the Design, Care and Use of Power Driven Brushing Tools.

National Stock Number	Outside Brush Diameter (Inches)	Wire Diameter (Inch)	Arbor Hole Diameter (Inches)	Wire Trim Length (min.) (Inches)	Wire Trim Length (max.) (Inches)	Number of Wire Points (min.)	Material	Max. Safe Free Speed (rpm)
5130-01-177-5609	3	0.0140	3/8	9/16	13/16	2600	CRS	20000
5130-00-405-9434	3	0.0204	1/2	9/16	13/16	1200	HTS	20000
5130-01-054-6779	3-1/8	0.0118	1/2	9/16	13/16	3400	CRS	20000
5130-01-028-5571	4	0.0104	1/2	3/4	1	4800	HTS	8000
5130-00-569-1402	4	0.0118	1/2	3/4	1	4800	HTS	8000
5130-01-023-4492	4	0.0140	3/8	3/4	1	2800	HTS	20000
5130-01-016-9020	4	0.0160	3/8	3/4	1	2500	HTS	12000
5130-00-277-9486	4	0.0160	1/2	3/4	1	2500	HTS	8000
5130-00-449-3799	6	0.0118	1/2	1-3/16	1-9/16	7200	HTS	6000
5130-00-596-0673	6	0.0118	5/8	1-3/16	1-9/16	7200	HTS	6000

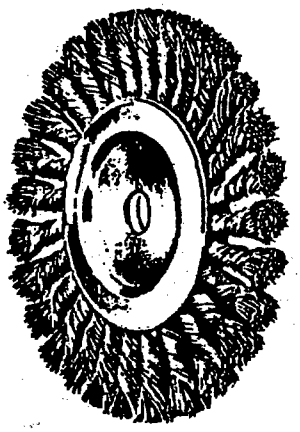
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National Stock Number	Outside Brush Diameter (Inches)	Wire Diameter (Inch)	Arbor Hole Diameter (Inches)	Wire Trim Length (min.) (Inches)	Wire Trim Length (max.) (Inches)	Number of Wire Points (min.)	Material	Max. Safe Free Speed (rpm)
5130-00-277-9487	6	0.0160	5/8	1-3/16	1-9/16	4600	HTS	6000
5130-00-293-0781	6	0.0230	5/8	1-3/16	1-9/16	2000	HTS	6000
5130-00-293-1983	6	0.0300	1/2	1-3/16	1-9/16	1500	HTS	6000
5130-00-066-0962	7	0.0140	5/8	1-9/16	1-15/16	6200	HTS	4500
5130-00-943-6986	8	0.0118	5/8	1-9/16	1-15/16	12000	HTS	4500
5130-01-022-4935	8	0.0160	5/8	1-9/16	1-15/16	6500	CRS	6000
5130-00-277-9488	8	0.0160	5/8	1-9/16	1-15/16	6500	HTS	4500
5130-00-293-2713	10	0.0118	1-1/4	1-3/4	2-1/2	17000	HTS	4250
5130-01-054-6777	12	0.0160	1-1/4	2-3/4	3-7/16	10500	CRS	3600
5130-00-277-9485	12	0.0204	1-1/4	2-3/4	3-7/16	6400	HTS	3600

CRS - corrosion-resisting steel
HTS - high carbon tempered steel

Tolerance: Outside diameter - plus or minus 1/8 inch.
Wire diameter - plus or minus 0.001 inch.
Arbor hole diameter - for 5/8 inch and smaller, plus 0.005 inch, minus 0 inch.
for larger than 5/8 inch, plus 0.006 inch, minus 0 inch.



Free speed test. The brush shall be attached to a power source and rotated as follows:

under 6 inch brush diameter, rotated at a speed 20 percent faster than the maximum free speed indicated in the table.
between 6 and 8 inch, inclusive, brush diameter, rotated at a speed 30 percent faster than the maximum free speed indicated in the table.
over 8 inch brush diameter, rotated at a speed 50 percent faster than the maximum free speed indicated in the table.

The brush shall be shrouded in an enclosure capable of capturing all fill wire or other debris that may become separated from the brush during the test. The brush shall be rotated at the maximum free speed indicated in the table for one minute. The interior of the enclosure shall then be examined for any wire or other foreign matter ejected from the brush. Any evidence of fill wire separation, other than trimmings, shall be cause for rejection of the lot.

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Quality assurance provisions.

Inspection. Sampling for inspection shall be in accordance with MIL-STD-105, Sampling Procedures and Tables for Inspection by Attributes, except where otherwise indicated herein. The lot size shall consist of all brushes of the same style and size offered for acceptance at one time. The sample unit is one brush.

Examination for visual requirements. Each brush shall be examined for workmanship and marking requirements. The inspection level shall be Inspection Level II with an acceptable quality level (AQL) of 2.5 defects per hundred units.

Examination for dimensional requirements. Each brush shall be examined for dimensional requirements. The inspection level shall be Inspection Level I with an AQL of 2.5 defects per hundred units.

Testing for free speed test requirements. Each brush shall be examined for conformance to the requirements of the free speed test. The inspection level shall be Special Inspection Level S-3 with an AQL of 2.5 defects per hundred units.

Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

NOTES:

The issue of the referenced documents in effect on the date of the solicitation shall be used to determine compliance with the stated requirements.

ANSI Standards are available from American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.

Military Standards are available from Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

PREPARING ACTIVITY:

GSA-FSS

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